



PRODUCT NAME

Pet Dental Care Formulation

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Short Description

A natural dental care powder designed to freshen breath and reduce plaque build-up when added to pet food. Contains finely ground charcoal and peppermint for effective oral hygiene. Suitable for both dogs and cats.

Technical Specifications

pH Level:	Neutral (approx. 7.0)
Viscosity:	Powder form
Shelf Life:	12 months
Batch Size:	Variable, based on demand and equipment capacity
Processing Time:	Approximately 2 hours
Temperature:	Room temperature
Storage Conditions:	Store in a cool, dry place away from moisture and direct sunlight.

Equipment:	Mixing bowls, accurate weighing scales, packaging containers
Certification:	ISO 9001, GMP compliant

Formulation Table

Ingredient	INCI Name	%	Function
Bentonite Clay	Bentonite	25.0%	Binder and cleanser; helps remove toxins and debris
Activated Charcoal	Activated Charcoal Powder	20.0%	Natural whitener and cleanser
Calcium Carbonate	Calcium Carbonate	20.0%	Mild abrasive to clean teeth and neutralize acids
Peppermint Extract	Mentha Piperita (Peppermint) Extract	10.0%	Natural breath freshener
Diatomaceous Earth	Diatomaceous Earth	15.0%	Mild abrasive to remove plaque
Sodium Bicarbonate	Sodium Bicarbonate	10.0%	Deodorizer and pH balancer

Manufacturing Process

Phase 1: Preparation Phase

1. Measure all ingredients accurately.
2. Ensure all equipment and tools are sanitized and ready for use.

Phase 2: Mixing Phase

1. In a large mixing bowl, combine Bentonite Clay, Activated Charcoal, and Calcium Carbonate.
2. Gradually add Diatomaceous Earth and Sodium Bicarbonate while mixing.
3. Ensure an even distribution of all powders throughout the mixture.

Phase 3: Addition Phase

1. Add Peppermint Extract to the powder blend.
2. Mix thoroughly to ensure uniform flavor distribution.

Phase 4: Quality Control and Packaging

1. Check the homogeneity of the product.
2. Store in airtight containers to prevent moisture absorption.

Required Equipment

Mixing bowls, accurate weighing scales, packaging containers

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, and lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Store in a cool, dry place away from moisture and direct sunlight.

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: ISO 9001, GMP compliant

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for Variable, based on demand and equipment capacity.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.